

Set-theoretic estimation at scale: 86 documents through the chunking pipeline — The Federalist Papers

Haiku/Sonnet extracted the frames; Lean ran the proved algorithm

thejeb project note

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Abstract

We scale the empirical STE pipeline from seven short passages (the tides note) to a large corpus of long documents: the **86 essays of The Federalist Papers**. Each essay is one STE “voice.” A general, topic-parameterized pipeline *chunks* every document, extracts open frames per chunk with **Claude Haiku**, unifies a canonical 10-variable schema with **Claude Sonnet**, and re-extracts one frame per document against that schema; the 86 frames are then scored by the *verified* computable STE layer (`Ste.FiniteInstance`) running in a Lean executable that reads the frames at runtime. The verdict, computed by the proved algorithm: the corpus is *inconsistent*, but the inconsistency is *narrow* — 7 of 10 variables are unanimous where spoken (a large consistent core), and exactly **3 conflict sites** carry the whole obstruction, headlined by a genuine three-way split on how Publius characterizes the proposed government. The machine recovered the real, well-documented internal tension of The Federalist.

1 The corpus and the pipeline

The corpus is the 86 essays of The Federalist [Hamilton et al., 1788] (85 numbers; No. 70 appears twice in the source edition), split from Project Gutenberg eBook #18 into one public-domain file per essay with a provenance header. The median essay is $\approx 13\text{k}$ characters (max $\approx 34\text{k}$) — long enough that per-document *chunking* is required, not optional.

The pipeline (`scripts/corpus/`) is the tides pipeline generalized to any corpus and topic, with chunking and parallel model calls:

1. **Pass 1 — Haiku, open, per chunk.** Each document is split into $\approx 9\text{k}$ -char chunks on paragraph boundaries; each chunk is extracted independently into (variable, value, quote) claims, unioned per document. 86 documents $\rightarrow$ **189 chunks**.
2. **Pass 2 — Sonnet, unify, once.** The 189 chunks proposed **2,510 distinct** (variable, value) pairs; Sonnet merged them into a canonical **10-variable** schema with closed domains and a synonym map.
3. **Pass 3 — Haiku, schema-constrained, per document.** Each of the 86 documents is re-extracted against the canonical schema into one frame (a domain value or silent per variable, with a licensing quote).

The run made ≈ 276 model calls ($189 + 1 + 86$) in parallel (`--jobs 8`) in roughly half an hour. The output `frames.json` has exactly the shape the verified Lean executable reads.

2 The verdict — the proved algorithm on runtime data

lake exe tides frames.json reads the 86 frames and evaluates the *verified* STE.Consistent and STE.disagreementDegree of Ste.FiniteInstance — the definitions proved (in that module) to characterize the Set-level feasibility of Ste.Basic (feasibilitySet_eq_empty_iff). No native_decide; the aggregation logic is the proved one, run on the loaded data.

Variable	degree	reading
ratification_stance	1	unanimous (supports ratification)
union_vs_disunion	1	unanimous (single union essential)
standing_army_restriction	1	unanimous
executive_structure	1	unanimous (single executive)
judicial_power	1	unanimous (safe, essential check)
bill_of_rights_necessity	1	unanimous
extended_republic_theory	1	unanimous (extended republic superior)
federal_power_scope	2	32 extensive vs. 13 strictly-limited
senate_equal_suffrage	2	5 justified vs. 1 unjust
government_character	3	mixed / federal / national

- **Inconsistent** (STE.Consistent = false): the feasibility set is empty — 86 essays by one pseudonymous author do not admit a single joint reading.
- **But narrowly so. 7 of 10 variables are unanimous where spoken** (disagreement degree ≤ 1): ratification, union, the standing army, the executive, the judiciary, the bill of rights, and the extended republic. That is a large *consistent core* — the shared Federalist program.
- **Three conflict sites** carry the whole obstruction. The headline is government_character, a genuine *three-way* split (degree 3): mixed_composite (25 essays, incl. Federalist 39’s “neither a national nor a federal Constitution, but a composition of both”), federal_confederacy (11 essays, e.g. Federalist 45’s “powers delegated... few and defined”), and national_consolidated (Hamilton’s Federalist 12/15). The other two are federal_power_scope (extensive vs. strictly limited) and senate_equal_suffrage.

The pipeline recovered, from raw text and by a proved decision procedure, the real internal tension long noted by readers of The Federalist: Publius describes the proposed government three different ways depending on rhetorical need, and the essays differ on the reach of federal power.

3 Contrast with the tides

	Tides (7 voices)	Federalist (86 voices)
Consistent?	no	no
Structure	pervasive, multi-camp	consistent core + 3 conflict sites
Worst variable	mechanism (degree 5)	government_character (degree 3)
Reading	rival theories	one program, few internal tensions

Same verified algorithm; two qualitatively different disagreement shapes. That the verdict is *computed* rather than asserted is exactly what lets it distinguish them.

4 The boundary of trust

The Lean layer certifies the aggregation, not the reading of the text. Each asserted value in `frames.json` carries the source sentence that licenses it, so every cell is auditable against the essay; the three conflict sites were spot-checked against quotes and are faithful. A mis-extraction would move a count, but the verdict itself — feasibility and the disagreement degrees — is the output of the proved algorithm on the frames, not a plausible story about them.

References

Alexander Hamilton, James Madison, and John Jay. *The Federalist Papers*. Independent Journal / New York Packet, 1788. 85 essays defending the proposed U.S. Constitution; public-domain text, Project Gutenberg eBook #18.